

Table 1. **Detailed comparison of a representative 31 of the 159 benchmarks** in the landscape (Table ??), grouped by evaluation lane. **Eval:** open open-loop prediction/generation quality, closed closed-loop task success, bridge prediction-to-action bridge. **VLA vs WM:** ✔ native direct-policy vs world-model contrast, ~ partial, -- model-agnostic. Venues are web-verified; “arXiv” marks preprint-only works.

Benchmark	Year	Venue	Eval	Capability	VLA vs WM
§3.1 Policy / manipulation suites					closed
CALVIN [18]	2021	RA-L	closed	long-horizon language	--
LIBERO [17]	2023	NeurIPS D&B	closed	short-horizon + transfer	--
Meta-World [28]	2020	CoRL	closed	multi-task / meta-RL	--
ManiSkill2 [10]	2023	ICLR	closed	generalizable skills	--
SIMPLER [16]	2024	CoRL	closed	generalization (real vs sim)	--
THE COLOSSEUM [23]	2024	RSS	closed	generalization / robustness	--
RoboCasa [20]	2024	RSS	closed	long-horizon (data scaling)	--
RoboArena [1]	2025	CoRL	closed	real-world generalization	--
GemBench [9]	2024	ICRA	closed	generalization levels	--
VLABench [30]	2025	ICCV	closed	long-horizon reasoning	--
§3.2 Embodied navigation, planning & social					closed
Habitat 3.0 [22]	2023	ICLR	closed	social / multi-agent	--
BEHAVIOR-1K [14]	2024	CoRL	closed	long-horizon household	--
ALFRED [26]	2019	CVPR	closed	long-horizon, partial-obs	--
TEACH [21]	2021	AAAI	closed	hidden-state (dialog)	--
RoboTHOR [7]	2020	CVPR	closed	navigation / sim-to-real	--
SocNavBench [3]	2021	ACM THRI	closed	social navigation	--
LIBERO-Mem [6]	2025	AAAI	closed	hidden-state / occlusion	✔
VIMA-Bench [13]	2023	ICML	closed	multimodal-prompt generalization	--
§3.3 World-model & video-generation evaluation					open
WorldModelBench [15]	2025	arXiv	open	video-WM prediction quality	--
Physics-IQ [19]	2025	WACV	open	physical realism	--
VBench-2.0 [31]	2025	arXiv	open	generation faithfulness	--
WorldScore [8]	2025	ICCV	open	world-gen quality	--
EWMBench [11]	2025	arXiv	open	scene/motion/semantic quality	--
EVA-Bench [5]	2024	ICML	open	embodied video anticipation	--
WorldPrediction [4]	2025	arXiv	open	counterfactual action recognition	--
Physion [2]	2021	NeurIPS D&B	open	physical prediction from vision	--
Physion++ [27]	2023	NeurIPS D&B	open	physical prediction with online property inference	--
§3.4 Bridges (prediction to action)					bridge
WorldSimBench [24]	2024	ICML	bridge	video-to-action consistency	~
World-in-World [29]	2025	ICLR	bridge	closed-loop task success of WMs	~
RoboWM-Bench [12]	2026	arXiv	bridge	prediction executability	~
WorldArena [25]	2026	arXiv	bridge	closed-loop WM arena	~